MILITARY SPECIFICATION

RECEIVER-TRANSMITTER, RADIO RT-542(*)/ASQ

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall provide two-way radio communication with properly equipped surface stations and other aircraft by means of amplitude modulated radio telephony. When installed with the necessary associated equipment the radio set shall provide:

(1) Communications by means of AM radio telephony (A3).

(2) Reception of special-type signals consistent with the performance requirements outlined herein.

(3) Automatic direction finding when installed with certain equipment.

(4) Automatic relay wherein reception of signals on one equipment will cause transmission on another as outlined herein.

1.2 Associated Equipment - The equipment shall operate with the following associated equipment which shall not be supplied as part of the equipment:

(1) Antenna AT-14A/A with RG-8/U cable

(2) Headset H-1/AR, H-4/AR, H-173/AR or equivalent

(3) Microphone - T-17, NAF-213264-6 or ANB-M-C1, carbon type

(4) Throttle Switch - NAF-1124-17

(5) Control, Radio Set C-2791/ARC or C-2790/ARC

(6) Intercommunication Set AN/AIC-14

(7) Junction Box J-17A/ARC-5

(8) Jack Box J-22/ARC-5

(9) Direction Finder Group AN/ARA-48

(10) Mounting-Plenum MT-2243/ASQ-56 or MT-2244/ASQ

(11) Amplifier-Power Supply-Receiver AM-2310/ASQ

FSC 5821

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2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-I-22668	Integrated Electronic Centrals AN/ASQ-56, AN/ASA-57 and AN/ASQ-58

STANDARDS

Military

MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of

Federal

FED STD 595	Colors (collection of 358 standards colors for paints and other materials)

2.2 Availability of Documents - When requesting specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code CDS, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, parts and materials requirements shall conform to Specification MIL-I-22668.

3.2 Design and Construction -

3.2.1 General - The equipment shall conform with all the applicable requirements of Specification MIL-I-22668 for design, construction and workmanship except in the paragraphs as modified below:

3.2.4.1 *Operational Stability* - The equipment shall operate with optimum performance for 200 hours continuously or intermittently without necessity for readjustment of any controls which are non-accessible during flight.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.5.1 Cables and Connectors - Connectors used for input wiring or interconnection of removable subassemblies shall be type DA manufactured by the Cannon Electric Company or an approved equivalent. External connectors shall be Amphenol Type 126-10572 or an approved equivalent. The wires and terminals of this connector shall be potted with a sealing compound which is capable of withstanding temperatures to 125°C.

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3.2.11 Standard Conditions - Cooling Air 25 lb/hr of ambient room air

3.2.12 Maximum Altitude - The maximum altitude under service conditions for the equipment shall be 75,000 feet.

3.2.12.1 Minimum External Cooling Requirements - The curve of Figure 1 specifies the minimum cooling air flow rates that are required for air inlet temperature varying from 0°F to 140°F in a 168°F environment. The cooling air weight flow as a function of the Radio Receiver-Transmitter pressure drop is specified in the following equation:

$$W = \left[1.09\ (\sigma \Delta P)\ 0.667\right] \pm 10\%\ \text{lb/min}$$

W = Cooling air flow rate (lb/min)

$\sigma \Delta P$ = Corrected total pressure (inches H_2O)

σ = Density correction P/P_r

P_r = Standard density (0.0765 lb/ft^3)

3.2.14 Input Power Requirements - The equipment shall meet all applicable requirements of MIL-STD-704 and shall give specified performance from the following power sources with characteristics as defined in MIL-STD-704, where applicable, having limits as specified therein. The power required shall not exceed the specified amounts.

(1) A. C. Power (Three Phase), Category "B"

Phase 1	150 va
Phase 2	12 va
Phase 3	12 va

(2) 130 V DC

Operating Voltage Limits	123 to 137 volts
Reduced Performance Voltage Limits	115 to 123 and 137 to 145 volts
Power, Maximum Operating Limits	48 watts

(3) 430 V DC

Operating Voltage Limits	404 to 453 volts
Reduced Performance Voltage Limits	382 to 404 and 453 to 464 volts
Power, Maximum Operating Limits	136 watts

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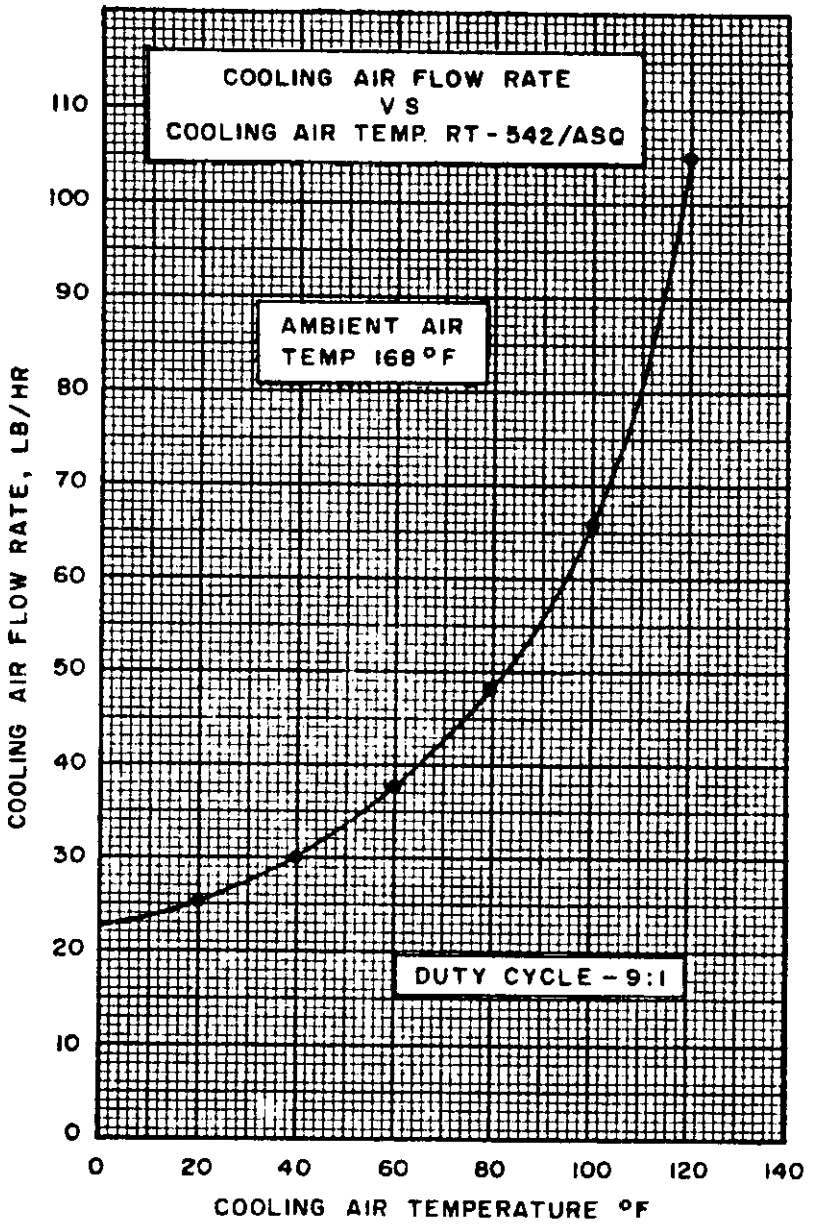


Figure 1

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(4) D. C. Power, Category "B"

Power, Maximum Operating Limits	50 watts
Power, Maximum during Channel Change	200 watts

3.3 Performance Requirements - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements apply under standard conditions to both main and guard channel operation. Reduced performance and service requirements shall apply when specifically outlined. Equipment performance under service conditions shall consist of a duty cycle wherein the equipment is operated on a continuous cycle of 10 minutes reception and 5 minutes transmission. At minimum operating temperature the equipment need not meet the requirements under service conditions until it has operated fifteen minutes.

3.3.1 Frequency Channels - By means of control box selection, it shall be possible to select any frequency in the range of 225.0 to 399.9 mc in 0.1 mc increments. These main channel frequencies shall correspond with the arithmetic series 225.00, 225.10, 225.20, 225.30 ----399.90 mc. Additionally, it shall be possible to select a fixed tuned guard channel in the range of 238.0 mc to 248.0 mc. Unless otherwise specified, the equipment shall be supplied with the guard frequency set for 243.0 mc.

3.3.2 Sensitivity - The sensitivity of the receiver shall be such that when an r-f input of 5 microvolts (open-circuit) modulated 30% at 1000 cps is applied the audio output power shall be at least 50 milliwatts with a signal plus noise-to-noise ratio (modulated to unmodulated) of 10 db or greater. When the main receiver is tuned to frequencies within 5 mc of the guard channel frequency, the main receiver and guard receiver shall meet the sensitivity requirement with 8 microvolts input. Under service conditions the sensitivity shall not be worse than 20 microvolts at any frequency. Under special service conditions of 22 volts dc and 102 volts ac power, the signal plus noise-to-noise ratio shall be at least 10 db for 10,000-microvolt signal modulated 30% at 1000 cps.

3.3.2.1 Degraded or Unuseable Channels Due to Internal Interference - Certain main receiver channel frequencies may be degraded or unuseable due to internally generated interference from the guard channel oscillator. When the equipment is supplied with the guard frequency set for 243.0 mc, the main receiver sensitivity measurements shall be made with an input of 20 microvolts at frequencies of 241.1, 241.2 and 275.6 mc. Two unuseable channels, 246.6 mc and 253.3 mc, are permitted.

3.3.3 AVC Characteristics - The AVC action shall be such that when a 1000-microvolt signal modulated at 1000 cps is applied, the audio output power shall be 25 milliwatts (±15%), and when the signal input (30% modulated at 1000 cps) is varied from 10 microvolts to 100,000 microvolts, the audio output power shall not vary by more than ± 3 db from the audio output obtained with the 1000-microvolt signal. When a main frequency within 5 mc of the guard frequency is selected, the foregoing requirement need be met only over the range 20 to 100,000 microvolts. Additionally, the receiver shall be protected against blocking under conditions of input signals up to one-half volt. The equipment shall be provided with an internal volume control which shall be adjusted so as to obtain the required output with the 1000-microvolt signal. Blocking shall be defined as a reduction in audio output to a level below 50 milliwatts.

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3.3.4 Selectivity - The over-all selectivity of the receivers shall fall within the limits set forth below for either standard or service conditions.

Attenuation Below	Total Bandwidth	
Resonant Response	Guard Receiver	Main Receiver
6 db	60 kc min.	60 kc min.
60 db	400 kc max.	200 kc max.

3.3.5 Images and Other Spurious Responses - When receiving on any channel the receiver shall attenuate images and other spurious responses to the maximum practicable extent and in any case not less than 60 db relative to the desired signal, except that the guard receiver shall be permitted three spurious frequencies attenuated less than 60 db but not less than 40 db.

3.3.6 I-F Rejection - Under either standard or service conditions, the receiver shall present an attenuation of not less than 80 db, relative to the response of the desired signal to signals on the frequencies of the intermediate frequency amplifiers.

3.3.7 Regeneration - The interstage and over-all regeneration of the receiver shall be so minimized that no evidence of unnecessary oscillation, instability, or excessive regeneration shall occur at maximum r-f gain setting under either standard or service conditions.

3.3.8 Noise Peak Limiter - The second detector of the receiver (following the intermediate frequency amplifier) shall be provided with an instantaneous noise peak limiter. The noise limiting action shall effectively limit the audio output voltage of modulating surges or peaks exceeding 50% +15% modulation of the r-f signal. Its action shall be rapid enough to prevent loss of intelligibility due to reduction in receiver gain under the influence of continuous r-f noises such as are caused by commutators, ignition systems, etc. When a pulsed signal having a pulse duration of 10 microseconds, a pulse repetition frequency of approximately 1000 pulses per second, and peak amplitudes up to 0.1 volt is applied on the same frequency as a desired signal modulated 30% at approximately 1000 cps, not more than 50 microvolts input (desired signal) shall be required to obtain a signal plus noise-to-noise ratio of 3 db.

3.3.9 Audio Characteristics - The audio characteristics of the receiver shall be as set forth below. Except where otherwise stated, these requirements apply under both standard and service conditions, and apply to signals detected by either the fixed frequency or variable frequency receivers and amplified in the common headset circuit.

3.3.9.1 Capability - Under standard conditions, the receiver shall be capable of delivering an output of at least 250 milliwatts when a signal having any level between 100 and 100,000 microvolts modulated 30% at 1000 cps is applied. With a 1000 microvolt input signal modulated 30% at 1000 cps the total harmonic content of the receiver output shall not exceed 10% at 250-milliwatts. Under service conditions the output shall not be more than 6 db below that obtained under standard conditions. Under the special service conditions wherein the d-c supply voltage is 22 volts dc and the a-c supply voltage is 102 volts ac, an output of at least 30 milliwatts shall be obtained when a 10,000-microvolt signal modulated 30% at 1000 cps is applied.

3.3.9.2 Over-all Fidelity - The audio response shall not vary more than +1 or -3 db at 300 cps, and +1 or -4 db at 4000 cps relative to the response at 1000 cps. Modulation frequencies outside the 300-4000 cps range shall be attenuated the maximum practicable without recourse to inclusion in the receiver of large and/or heavy filter components.

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3.3.9.3 Auxiliary Circuit - The auxiliary circuits shall deliver only signals detected by the variable frequency receiver. When a 1000-microvolt signal modulated 30% at 1000 cps is applied, the output shall be at least 0.25 volt, and when the modulation frequency is varied from 70 to 7000 cps, the output for any frequency in the range shall be within +2 and -3 db of the output obtained at 1000 cps.

3.3.10 Signal-to-Noise Type Squelch Characteristics - The main receiver squelch shall be controlled by the audio signal plus noise-to-noise ratio and shall be used with the r-f gain control set for maximum gain. Under conditions of 5 microvolts r-f input modulated at 400 cps, the squelch control provided within the receiver-transmitter shall have a range as to permit squelch operation from no modulation to greater than 30% modulation. With the squelch control adjusted to obtain normal audio output with modulation input required for 6 db signal plus noise-to-noise ratio, the audio shall be effectively suppressed when the modulation input is reduced to a 1 db signal plus noise to noise ratio.

3.3.10.1 Carrier Type Squelch Circuits - The guard receiver squelch shall be carrier operated and the main receiver shall have alternate soldered squelch circuit linkages for carrier squelch control in equipment automatic relaying applications. For these carrier type squelch circuits, the sensitivity controls provided in the interior of the receiver-transmitter shall have such a range as to permit a variation in the sensitivity of the receivers from maximum sensitivity to a minimum value of not less than 10 microvolts. These controls together with suitable "squelch" circuits provided in the receivers shall permit adjustment so as to effectively suppress unwanted signals or noise in the receiver output under conditions of no, or very low (depending upon the setting of the controls) r-f input signals to the receivers. Under conditions of the r-f gain controls adjusted so as to obtain at least 50 milliwatts output with an r-f input of 6 microvolts, 30% modulated at 1000 cps, the audio output of the receiver shall be effectively suppressed in the absence of r-f input at the antenna. Selection of a main frequency within 5 mc of the guard frequency shall cause an increase not exceeding two to one in all the foregoing values of microvolts. Operation of the silencer circuit shall be smooth and positive, requiring at its threshold of operation a change of approximately 1 db in r-f input (30% modulated at 1000 cps) for effecting a change in audio output of 10 db or more.

3.3.11 R-F Power Output - The transmitter shall be capable of delivering the following values of radio frequency carrier energy within the range described below:

	225.0 to 399.9 mc
Standard Conditions	16 watts min., 20 watts average
Service Conditions	9 watts min.
Reduced Performance Test	500 mw

3.3.12 Modulation Characteristics - The transmitter shall meet the following performance requirements for modulation when operated under both standard and service conditions.

3.3.12.1 Radio Telephony (A-3)

3.3.12.1.1 Modulation - Under standard conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter not less than 80% when the input voltage at 1000 cps obtained from the dummy microphone (see 3.4.1.8.4) is adjusted to not more than 1 volt for the carbon mircophone and to not more than 25 millivolts for the dynamic microphone. Modulation shall be determined from oscilloscope presentation of the modulated carrier wave

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and from measurement of maximum and minimum voltages displayed on the oscilloscope. Modulation shall be calculated from the formula:

$$\frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100\%$$

The modulator of this equipment shall employ clipper or limiter provisions for clipping positive and negative peaks at a modulation level of not more than 95% with increased microphone input voltage. The modulator gain control shall be adjusted to provide modulation to the clipping level with 1 volt (open circuit) input for the carbon microphone and 25 millivolts (open circuit) input for the dynamic mircophone. With the input reduced 3 db below the clipping level, the distortion of 1000 cps modulation envelope shall not exceed 10%. At this input level the transmitter shall have the following fidelity characteristics relative to the output at 1000 cps:

Frequency, cps	Relative Output in db
150	-8 or lower
300	-3 to -7
600	+1 to -2
1,000	0 (ref. level)
3,000	+2 to -2
6,000	+2 to -2
10,000	+1 to -5

3.3.12.1.2 Modulation Under Service Conditions - Under service conditions the modulator shall be capable of modulating the r-f carrier output of the transmitter at least 70%. Distortion of the modulation envelope with the input 3 db below the clipping level shall not exceed 10%.

3.3.12.1.3 Microphone Current - The microphone energizing current measured with dummy microphone connected and the equipment operated from 27.5-volt d-c supply shall be 50 ±7 milliamperes d-c.

3.3.12.2 Tone Modulation - Not less than 70% nor more than 100% modulation of the r-f carrier shall occur when tone transmission is selected and the transmitter is keyed. The frequency of modulation shall be between 920 and 1120 cps.

3.3.13 Spurious Radiation - The magnitude of all radiated signals on other than the carrier frequency shall be 40 db or more below the carrier signal except that harmonic frequencies shall be attenuated not less than 55 db at 450 mc with a linear increase to 75 db at 525 mc. The attenuation above 525 mc shall be 75db or greater.

3.3.14 Frequency Stability - The frequency stability of the complete equipment after operation under either standard or service conditions for five minutes or more shall be such as to maintain the equipment frequencies within the limits shown below provided the temperature does not exceed 55°C:

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Transmitter carrier frequency tolerance	±10 kc
Fixed-frequency receiver resonant frequency tolerance	±13 kc
Variable-frequency receiver resonant frequency tolerance	±10 kc

3.3.15 Sidetone - Monitoring of the carrier modulation shall be effected via the receiver audio output channel. The modulation signals delivered to the receiver output while the transmission is in effect shall be obtained by rectifying a small portion of the carrier output rather than from the modulator directly. Under standard conditions the minimum sidetone power output from the receiver shall be approximately 250 milliwatts when the transmitter carrier is modulated 80% at 1000 cps. Under service conditions the output shall be not less than 75 milliwatts. The total harmonic content of the output shall not exceed 15%.

3.3.16 Noise Modulation - Under standard conditions, noise modulation of the transmitter carrier shall be at least 40 db below the maximum modulation obtainable, and the receiver signal plus noise-to-noise ratio shall be at least 30 db when a 10,000-microvolt signal modulated 30% at 1000 cps is applied and the total receiver output (signal plus noise) is adjusted to be 250 milliwatts.

3.3.17 Channel Changing Cycle - The time elapsing between the instant the channel selector switch is positioned to select a given channel and the instant the equipment is in an operative condition on the selected channel shall not be greater than six seconds under service conditions. This requirement shall hold for shifting the operating frequency of the equipment between any two available channels. During channel changing operations, the output of the receiver (headset circuit) shall be shorted or otherwise reduced to zero to indicate, through lack of sidetone, that the equipment is not in readiness for transmission should the operator begin speaking before the channel changing cycle is completed.

3.3.18 Frequency Modulation - Under the service conditions and with no modulating voltage applied to the transmitter and no r-f signal applied to the receiver, the peak frequency deviation arising from frequency modulation of the receiver conversion oscillator or the transmitter receiver shall not exceed 1.0 kc. Under the standard conditions, and with the emitted carrier modulated 80% at 1000 cps, the peak frequency deviation arising from frequency modulation of the transmitter shall not exceed 1.0 kc.

3.3.19 Transmit-Receive Interval - When shifting from transmission to reception, or vice versa, on the same channel, not more than 300 milliseconds shall elapse between the operation of the control switch and the time the equipment is stablized in the selected mode of operation in compliance with the performance requirements. This requirement shall be met under service conditions after fifteen minutes operation.

3.3.20 Accuracy of Automatic Tuning - When the same frequency channel is selected several times in succession, the transmitter r-f power output and the receiver audio power (with an input signal of 100 microvolts or greater modulated 30% at 1000 cps) shall not vary by more than 0.5 db from the power outputs obtained initially. This requirement shall be met under standard conditions provided such conditions are static.

3.3.21 Cable Length - Under the standard or service conditions the equipment shall operate as required by this specification when using control cables up to 25 feet in length fabricated from No. 22 AWG copper wire, or when using cables up to 110 feet in length fabricated from No. 16 AWG copper wire.

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3.3.22 Relaying - When used in an automatic relaying installation, the receiver-transmitter shall be connected for carbon microphone input and carrier type squelch. When two equipments are connected for automatic relaying, the performance shall be as follows:

3.3.22.1 Single-Channel Communication - When transmission and reception only, on either receiver-transmitter, is selected, performance shall be as required by this specification for one receiver-transmitter in a nonrelaying installation.

3.3.22.2 Relaying Operation - The application to the antenna circuit of either radio set of a signal on the receiver resonant frequency and of a level sufficient to fully open the receiver squelch circuit shall cause the other radio set to generate and emit a radio-frequency carrier wave of the rated power on the selected frequency, and removal of the actuating signal shall cause cessation of the generated carrier. When the actuating signal is 3 db or more above the minimum level at which relaying in accordance with the foregoing sentence is possible, the generated carrier shall reach the rated power level within 300 milliseconds after application of the actuating signal and the central shall be returned to standby condition within 300 milliseconds after removal of the actuating signal.

3.3.22.3 Relaying Modulation - When the equipment is relaying in accordance with 3.3.22.2 and amplitude modulation is applied to the actuating signal, the emitted carrier shall be modulated in an essentially similar manner, but with the peaks clipped by the clipper in the modulator. When a signal 3 db above the level required to actuate the squelch (with sensitivity setting for squelch operation at 5 microvolts) is applied and is modulated 80% at 1000 cps, the emitted carrier shall be modulated at least 70%.

3.3.22.4 Relaying Monitor - When the equipment is relaying as described above, the operator's headset of both equipments shall be connected so as to monitor all signals relayed in either direction. For individual transmission the operator of either equipment can disable the automatic relaying operation by pressing the push-to-talk switch. Under these conditions, the audio output may decrease not more than 6 db. Distortion and signal plus noise-to-noise ratio at the terminals of the headset circuit and r-f carrier output and modulation shall be essentially the same as those obtained when ordinary operation is selected.

3.3.22.5 Relay Squelch Change - It shall be possible by changing not more than seven connections on a suitable terminal board to shift the receiver squelch circuits from the signal-to-noise ratio type to the carrier operated type. The carrier operated type is to be used while relaying and the signal-to-noise ratio type is to be used for nonrelaying operation.

3.3.23 Automatic Direction Finding - When installed with certain associated equipment, the radio set shall be suitable for automatic direction finding. When used for this service the receiver-transmitter will receive radio frequency signals from the direction finding antenna, amplify and demodulate them, and deliver the low-frequency modulation components to the direction finder amplifying and indicating equipment. When the radio set is connected in an automatic relaying installation, the automatic direction finding function is available on one receiver-transmitter. When the relaying function is selected on Control, Radio Set C-2791/ARC, the adf function is disabled.

3.4 Detailed Requirements -

3.4.1 Receiver-Transmitter, Radio RT-542/ASQ - The receiver-transmitter shall meet the following requirements.

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3.4.1.1 Function - The receiver portion of the equipment shall receive r-f energy from the antenna, amplify and convert it to aural intelligence. The transmitter portion of the equipment shall generate r-f power and transmit it to the load specified herein.

3.4.1.2 Form Factor - The receiver-transmitter shall be contained in a single unit capable of being directly mounted to a shockmounted equipment shelf which in turn is mounted in an aircraft. It shall be possible to remove the unit from the equipment shelf and to remove the dust cover using ordinary hand tools without requiring removal of any cable. The over-all dimensions shall not exceed 7-17/32 inches high by 8-57/64 inches wide by 20-5/22 inches long.

3.4.1.3 Weight - The weight of the receiver-transmitter, less mounting, shall not exceed 44.5 pounds.

3.4.1.4 Contents - The equipment shall contain the complete transmitter, modulator, receiver, direct crystal-controlled frequency generator, frequency selection system, automatic tuning devices, and all other parts to make the equipment function as a unit for the associated equipment as specified herein. It shall be possible to remove or replace any plug-in unit located on the main chassis using ordinary hand tools and without unsoldering any connections. It shall be further possible to replace any plug-in units with another without resynchronization except that realignment of tuning elements may be necessary. The above circuits shall be contained in the following 11 modular assemblies.

(1) Amplifier-Preamplifier Module - Provides r-f amplification and first mixer during both receive and transmit modes.

(2) 20-30 MC I-F Amplifier Module - Provides first i-f amplification and second mixer during reception and first injection frequency during transmission.

(3) 1.85 MC I-F Amplifier Module - Provides second i-f amplification and detector circuitry.

(4) Audio Amplifier Module - Provides audio amplification.

(5) Spectrum Generator Module - Provides high frequency injection.

(6) Oscillator Module - Provides low frequency injection.

(7) Power Amplifier Module - Provides r-f amplification during transmission.

(8) Guard Receiver Module - Provides emergency receiver circuitry except for audio amplification.

(9) Relay Module - Provides power distribution circuitry.

(10) Modulator Module - Provides audio amplification and modulates the r-f carrier.

(11) Mechanical Drive Module - Tunes r-f, oscillator, and first i-f circuitry by correctly positioning the gears connecting these modules.

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3.4.1.5 Electrical Connection:

3.4.1.5.1 External

Reference Designation	Receptacle Type	Function
P-1401	Amphenol 126-10572 (or approved equivalent)	Power, control and r-f connections

3.4.1.5.2 Internal - The following electrical connections shall be accessible when the equipment is removed from its case:

(1) A Type JK-34, or an approved equivalent, jack for headset connection.

(2) A Type JK-33, or an approved equivalent, jack for microphone connection.

3.4.1.5.3 Test Connector - A test connector shall be provided on the front of the equipment for use with AN/USA-11 test equipment. This connector shall be type Bendix Pygmy PT076-10-6S or approved equivalent. The following functions shall be provided to the connector:

Function	Pin Letter
Main Audio	A
Aux Audio	B
T/R Relay Ground	C
Mike Input Hi	D
Ground	E
Mike Input Low	F

3.4.1.6 Controls, Internal - The following semi-fixed adjustable controls shall be accessible with the unit removed from its case:

(1) Main receiver sensitivity control.

(2) Guard receiver sensitivity control.

(3) Main audio amplifier gain control.

(4) Guard audio amplifier gain control.

(5) Sidetone audio gain control.

(6) Modulator audio gain control.

(7) Signal-to-noise squelch control.

(8) Auxiliary audio gain control.

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3.4.1.7 External Finish - The equipment covered by this specification shall be painted with color No. 2615 of FED-STD-595. The bottom surface of the equipment shall not be painted.

3.4.1.8 Terminations

3.4.1.8.1 Receiver R-F - R-f input to the receiver shall be through a 52-ohm cable from a signal generator having an effective source impedance of 52 ohms. Sensitivity shall be in terms of "open-circuit" microvolts.

3.4.1.8.2 Transmitter R-F - The r-f output load (dummy antenna) shall consist of a 52-ohm noninductive load which shall include means suitable for measuring r-f power output, percentage modulation and distortion.

3.4.1.8.3 Receiver Audio - The main (headset) audio channel of the receiver shall be terminated in a 300-ohm noninductive load which shall include means for measuring the audio-frequency power output. The auxiliary (wide-range) audio output channel of the receiver shall be terminated in a resistance load of 20,000 ohms, including means for measuring the audio output voltage.

3.4.1.8.4 Transmitter Input - The audio (microphone) input circuit of the transmitter shall be connected to an audio-frequency generator and dummy microphone having characteristics as listed below:

Carbon Microphone Input

Resistance - 82 ohms

Impedance (mainly resistive at audio frequencies) - 82 ohms

Frequency range - 150-10,000 cps

Total harmonic distortion - less than 1%

Open-circuit output potential (adjustable) 0.05-3.0 V rms

Dynamic Mike Input

Resistance - 82 ohms

Impedance (mainly resistive at audio frequencies) - 82 ohms

Frequency range - 150-10,000 cps

Total harmonic distortion - less than 1%

Open-circuit output potential (adjustable) - 1-50 mv rms

3.4.1.9 Pressurization - The case containing this equipment requires a supply of dry pressurized air. At an interval pressure of 20 psi absolute, the case construction shall be such that leakage is no greater than 5 psi in 24 hours when the external pressure is reduced to 0.65 psi absolute (70,000 feet). The equipment performance shall be in accordance with this specification when the internal pressure is maintained at 20 ± 5 psi absolute.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the test requirements of Specification MIL-I-22668 to determine compliance with all the applicable requirements except as noted herein.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-I-22668.

5.2 Final Adjustments - Unless otherwise stated by contract the receiver-transmitter shall be shipped properly adjusted for signal-to-noise squelch and carbon microphone operation.

6. NOTES

6.1 Use - This equipment is mainly intended to supply the uhf communication function for the Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57 and AN/ASQ-58.

6.2 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) System Specification - Integrated Electronic Centrals AN/ASQ-56, AN/ASQ-57 and AN/ASQ-58 Specification MIL-I-22668 shall have precedence over this specification.

(3) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the procuring activity.

(4) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 The parentheses (*), when used in the type designation, shall be replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with 3.2.10 of Specification MIL-I-22668. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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